Chapter 22

Introduction to Spectrochemical Methods

***Fundamentals of Analytical Chemistry*, Skoog, et al., 10th Ed., 2022**

Chapter 22 Problems:

22-2, 22-6, 22-7, 22-9, 22-11, 22-13, 22-14, 22-15, 22-19, 22-21

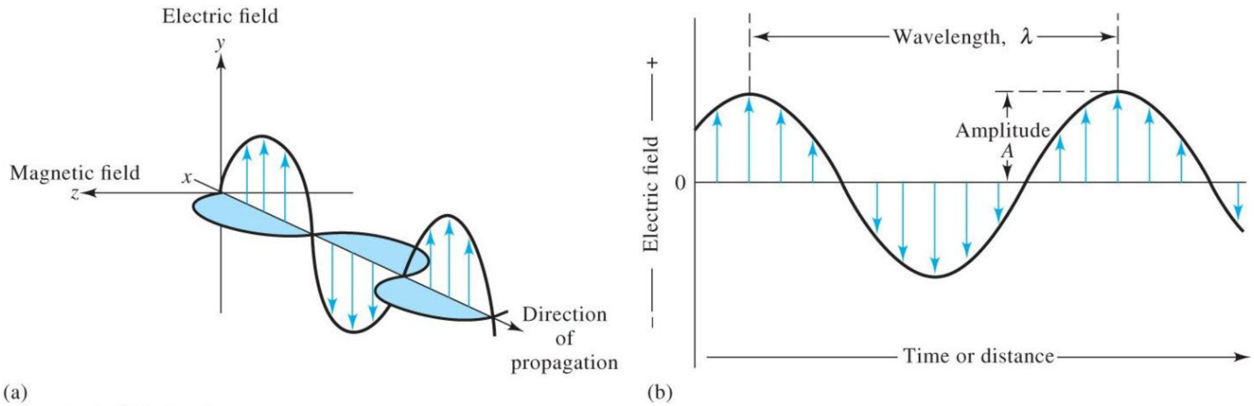
PROPERTIES OF ELECTROMAGNETIC RADIATION

Electromagnetic radiation can be described as a wave with properties of wavelength, frequency, velocity, and amplitude.

(light does not require a medium for transmission, unlike sound)

Electromagnetic radiation is can be treated as discrete packets of energy or particles called photons or quanta.

Wave Properties



The Amplitude of an electromagnetic wave is a vector quantity that provides a measure of the electric or magnetic strength at a maximum in the wave.

The Period of an electromagnetic wave is the time in seconds for a successive maxima or minima to pass a point in space.

The Frequency of an electromagnetic wave is the number of oscillations that occur in one second, units are Hz = sec^{-1} = cycles per second.

The wavelength is the linear distance between successive maxima (or minima)

The wavelength tells us what substance is there

TABLE 24-1

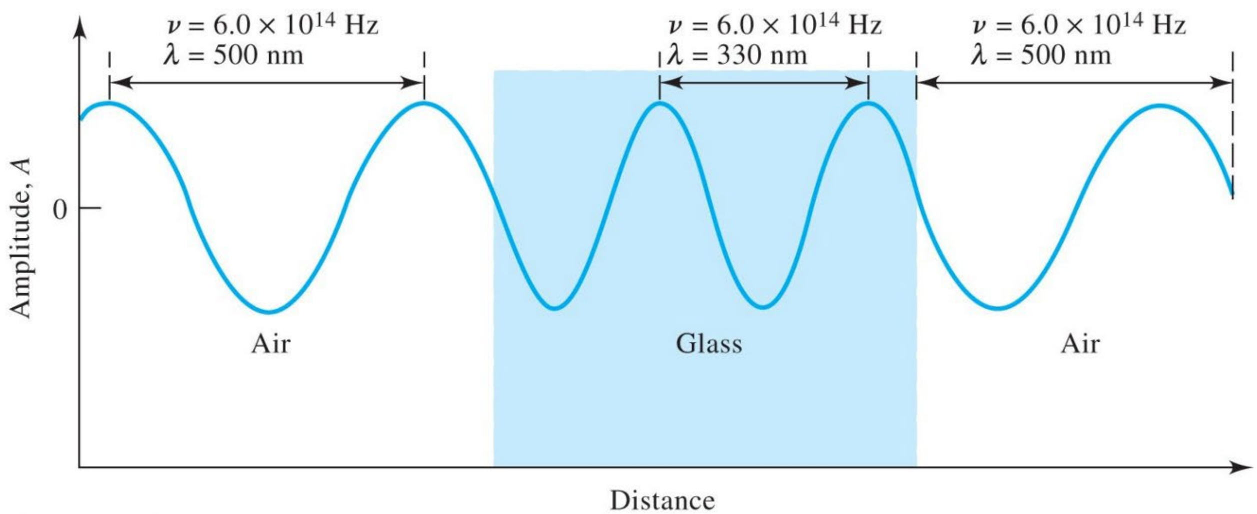
Wavelength Units for Various Spectral Regions

Region	Unit	Definition
X-ray	Angstrom unit, Å	10^{-10} m
Ultraviolet/visible	Nanometer, nm	10^{-9} m
Infrared	Micrometer, μm	10^{-6} m

Frequency is independent of the medium while velocity does change with changing medium.

Velocity = frequency x wavelength

$$\upsilon = \nu \ \lambda$$



The Speed of Light

$$c = \nu \ \lambda = 3.00 \times 10^{8} \text{ m/s} = 3.00 \times 10^{10} \text{ cm/s}$$

The wavenumber is another way to describe electromagnetic radiation. The wavenumber $\bar{\nu} = 1/\lambda$ it has units of cm^{-1}. ***The wavenumber is often used to describe radiation in the IR region.***

1. Calculate the wavenumber of a beam of infrared radiation with a wavelength of 5.00 μm.

Radiant Power and Intensity

The radiant Power P in watts W is the energy of a beam that reaches a given area per unit time.

$$Watts = \frac{J}{m^2\ s}$$

The Particle Nature of Light: Photons

The energy of a single photon to its wavelength, frequency and wavenumber

$$E = h\nu = \frac{hc}{\lambda} = hc\bar{\nu}$$

2. **Calculate the energy in joules of one photon of radiation with the a wavelength of 5.00 μm (or 2000 cm^{-1})**

Spectroscopic Measurements

Spectroscopists use the interactions of radiations with matter to obtain information about a sample

The sample is stimulated by applying energy in the form of heat, electrical energy, light particles, or a chemical reaction.

Initially the sample is predominately in its lowest energy state- called the ground state.

Stimulus cause some the analyte to undergo a transition to an excited state.

Information is obtained about the analyte by measuring the electromagnetic radiation that is absorbed or emitted as it returns to a ground state.

When a sample is stimulated several processes are possible

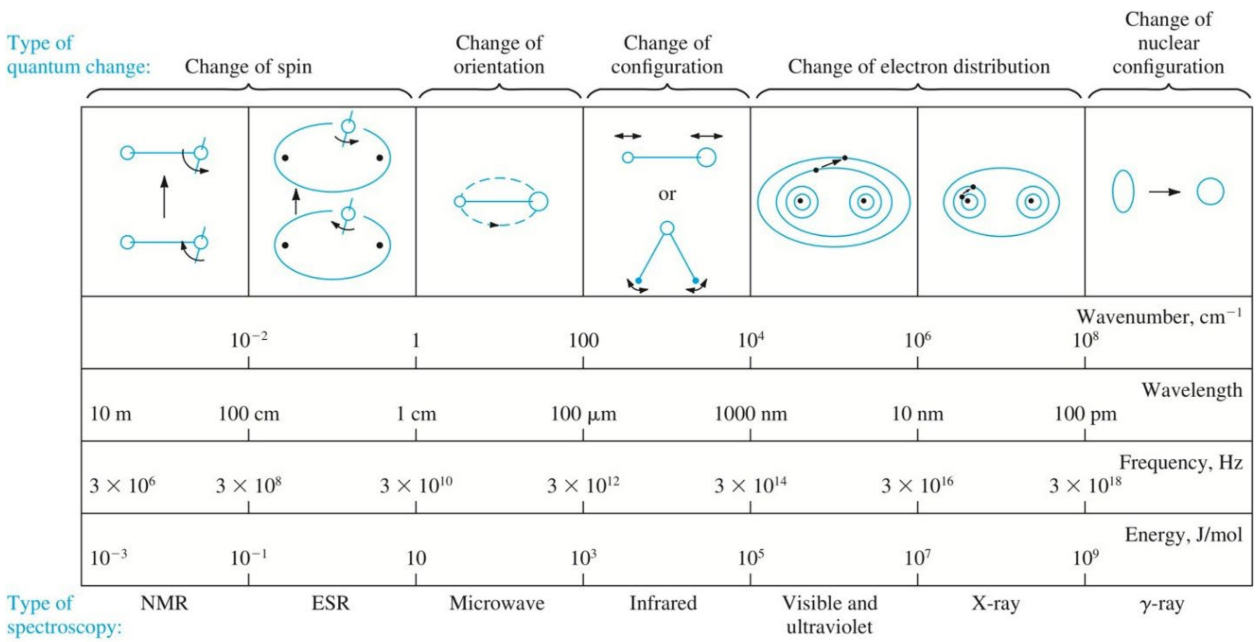
Radiation can be scattered, reflected, or absorbed

INTERACTION OF RADIATION AND MATTER

TABLE 24-2

Regions of the UV, Visible, and IR Spectrum

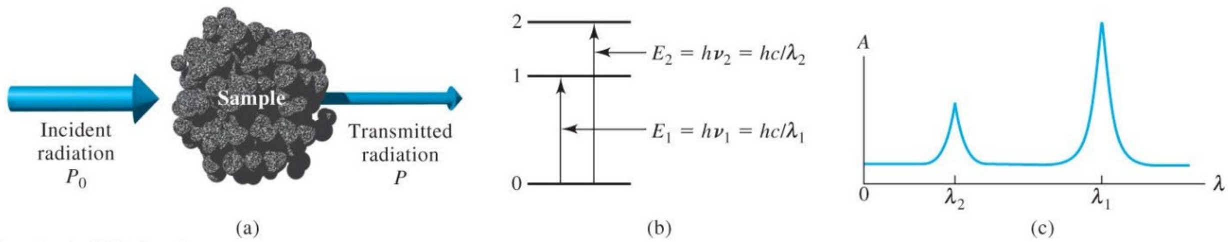
Region	Wavelength Range
UV	180–380 nm
Visible	380–780 nm
Near-IR	0.78–2.5 μm
Mid-IR	2.5–50 μm



(From C. N. Banwell, *Fundamentals of Molecular Spectroscopy*, 3rd ed., New York; McGraw-Hill, 1983, p. 7.)

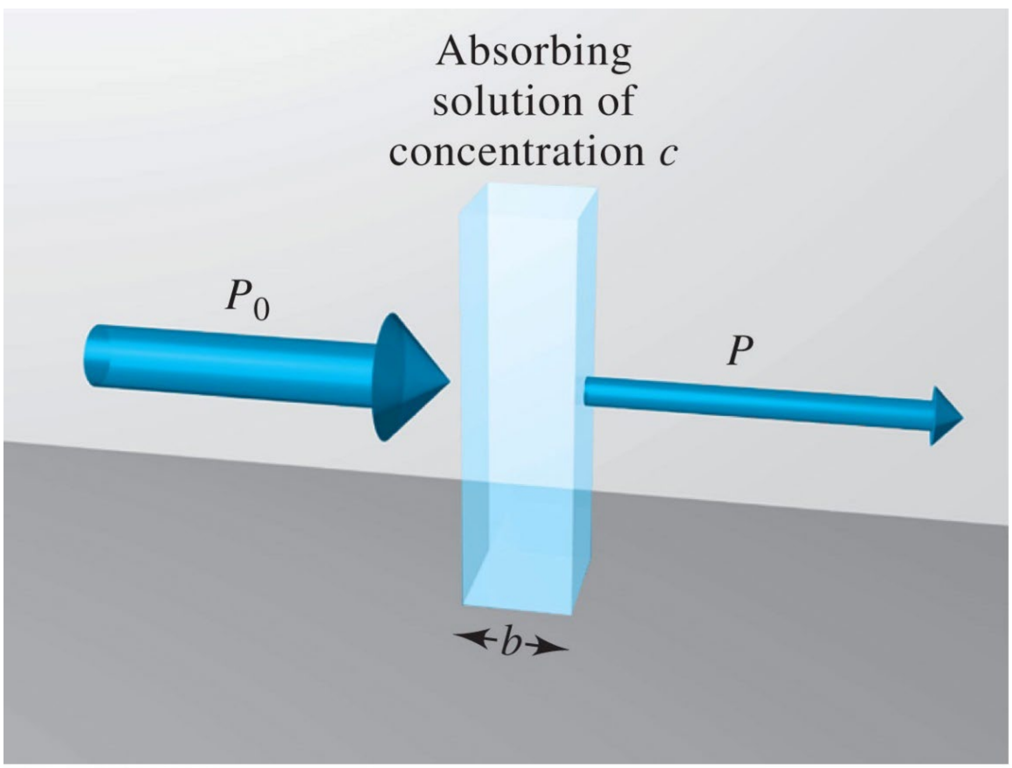
Absorption Spectroscopy

In absorption spectroscopy we measure the amount of light absorbed as a function of wavelength



ABSORPTION OF RADIATION

The Absorption Process



Transmittance T is the fraction of incident radiation transmitted by the solution

$$T = P/P_o$$

Percent Transmittance

$$\% T = P/P_o \times 100\%$$

Absorbance A is related to the transmittance in a logarithmic manner

$$A = -\log T$$

$$A = -\log P/P_o$$

$$A = \log P_o/P$$

$$A = -\log \%T + \log 100$$

$$A = 2 - \log \%T$$

If absorbance equals zero

$0 = 2 - \log \% T$

$2 = \log \% T$

Then $\% T = 100 \%$

If $\% T = 0 \%$

$A = 2 - \log 0$

Then $A = 2$

Beer Lambert Law (aka Beer's Law)

$$A = \log P_o/P = abc$$

c = concentration of absorbing species (g L^{-1})

b = pathlength (cm)

a = proportionality constant (units = L g^{-1} cm^{-1})

When we express the concentration molarity the proportionality constant is called the molar absorptivity ε (units = L mol^{-1} cm^{-1})

$$A = \varepsilon bc$$

3. A 7.25×10^{-5} M solution of potassium permanganate has a transmittance of 44.1% when measured in a 2.10 cm cell at a wavelength of 535 nm. Calculate

a) The absorbance of this solution

b) The molar absorptivity of $KMnO_4$

4. A solution containing 4.48 ppm $KMnO_4$ has a transmittance of 0.309 in a 1 cm cell at 520 nm. Calculate the molar absorptivity and the absorptivity of $KMnO_4$.

Performance Characteristics of Instruments

Figures of merit are characteristic performance criteria that are expressed in numerical terms.

TABLE 1-3 Numerical Criteria for Selecting Analytical Methods

Criterion	Figure of Merit
1. Precision	Absolute standard deviation, relative standard deviation, coefficient of variation, variance
2. Bias	Absolute systematic error, relative systematic error
3. Sensitivity	Calibration sensitivity, analytical sensitivity
4. Detection limit	Blank plus three times standard deviation of the blank
5. Dynamic range	Concentration limit of quantitation (LOQ) to concentration limit of linearity (LOL)
6. Selectivity	Coefficient of selectivity

TABLE 1-5 Figures of Merit for Precision of Analytical Methods

Terms	Definition*
Absolute standard deviation, s	$s = \sqrt{\frac{\sum_{i=1}^{N} (x_i - \bar{x})^2}{N - 1}}$
Relative standard deviation (RSD)	$\mathrm{RSD} = \frac{s}{\bar{x}}$
Standard error of the mean, s_m	$s_m = s/\sqrt{N}$
Coefficient of variation (CV)	$\mathrm{CV} = \frac{s}{\bar{x}} \times 100\%$
Variance	s^2

Sensitivity: Measure of the ability of an instrument to discriminate between small differences in analyte concentration.

Two factors limit sensitivity

Slope

Precision

If two methods have equal precision the one that has the steeper slope will be more sensitive.

<u>Calibration Sensitivity</u> = slope

<u>Analytical Sensitivity</u> takes into account precision

$\gamma = m/s_s$

m = slope

s_s = standard deviation of measurement (may vary with concentration)

<u>Detection Limit</u>

Minimum concentration or mass of analyte that can be detected at a known confidence level. This depends on the ratio of the signal to the size of the statistical fluctuation in the blank signal.

Detection Limit is the conc. that gives a signal that is equal to twice the peak-peak noise level of the baseline.

Detection Limit is really the lowest conc. that can be reliably detected.

Detection Limit is the analyte conc. that produces a signal that is 3 std. Dev. greater than the blank signal.

$S_m = 3xs_{bl} + S_{bl}$

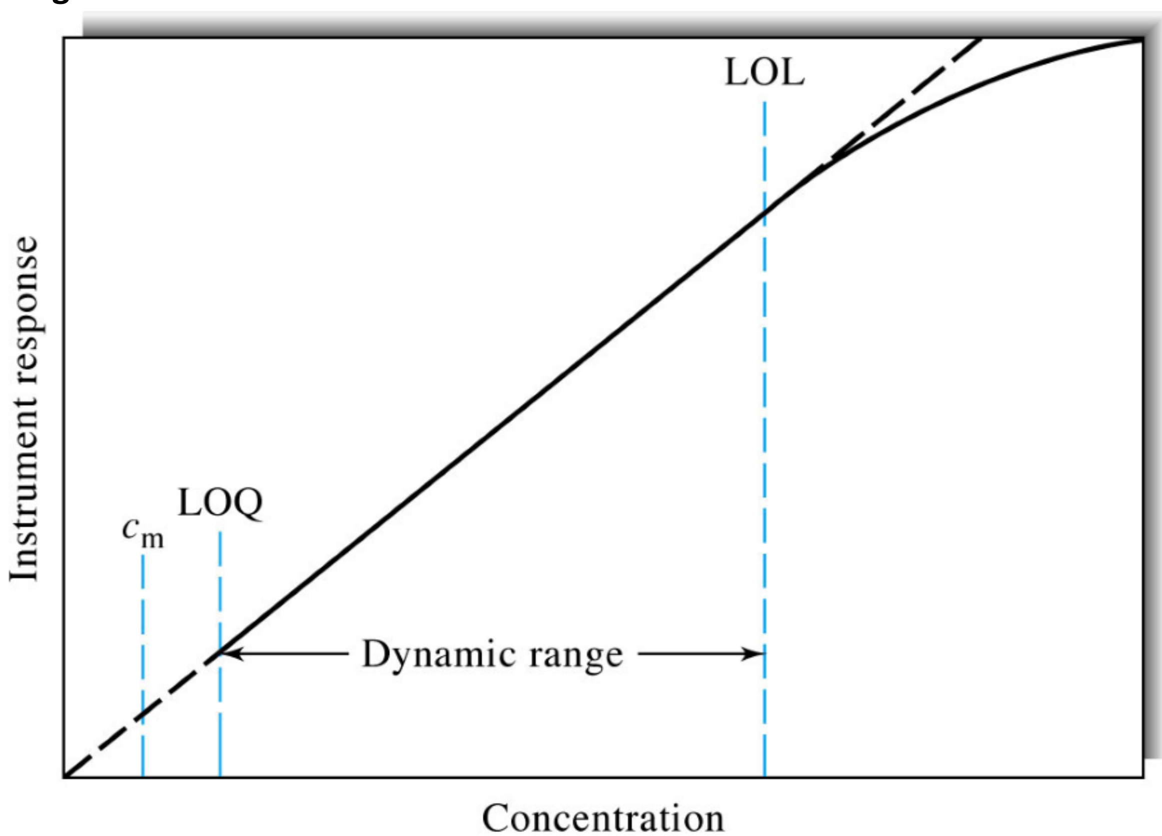
Standard concentrations must bracket the range of sample concentrations

Dynamic Range

Extends from the lowest concentration at which quantitative measurements can be made (limit of quantitation LOQ) to the concentration at which the calibration curve departs from linearity (limit of linearity LOL) usually by 5%.

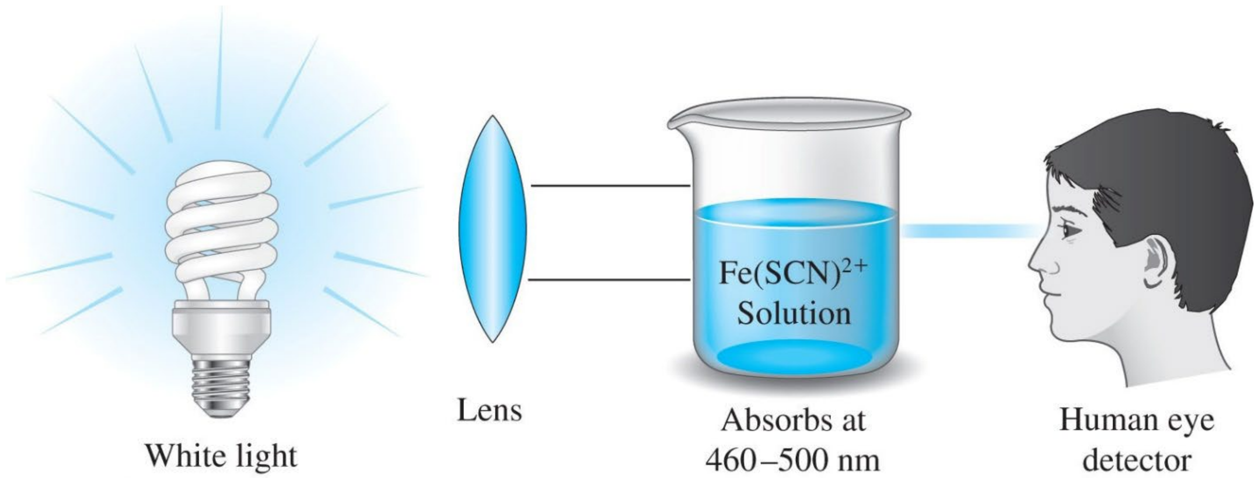
LOQ = 10 x standard deviation of the blanks

A useful method will have at least a few orders of magnitude dynamic range.



Standard concentrations must bracket the range of sample concentrations

Draw noise with signal vs time for LOD



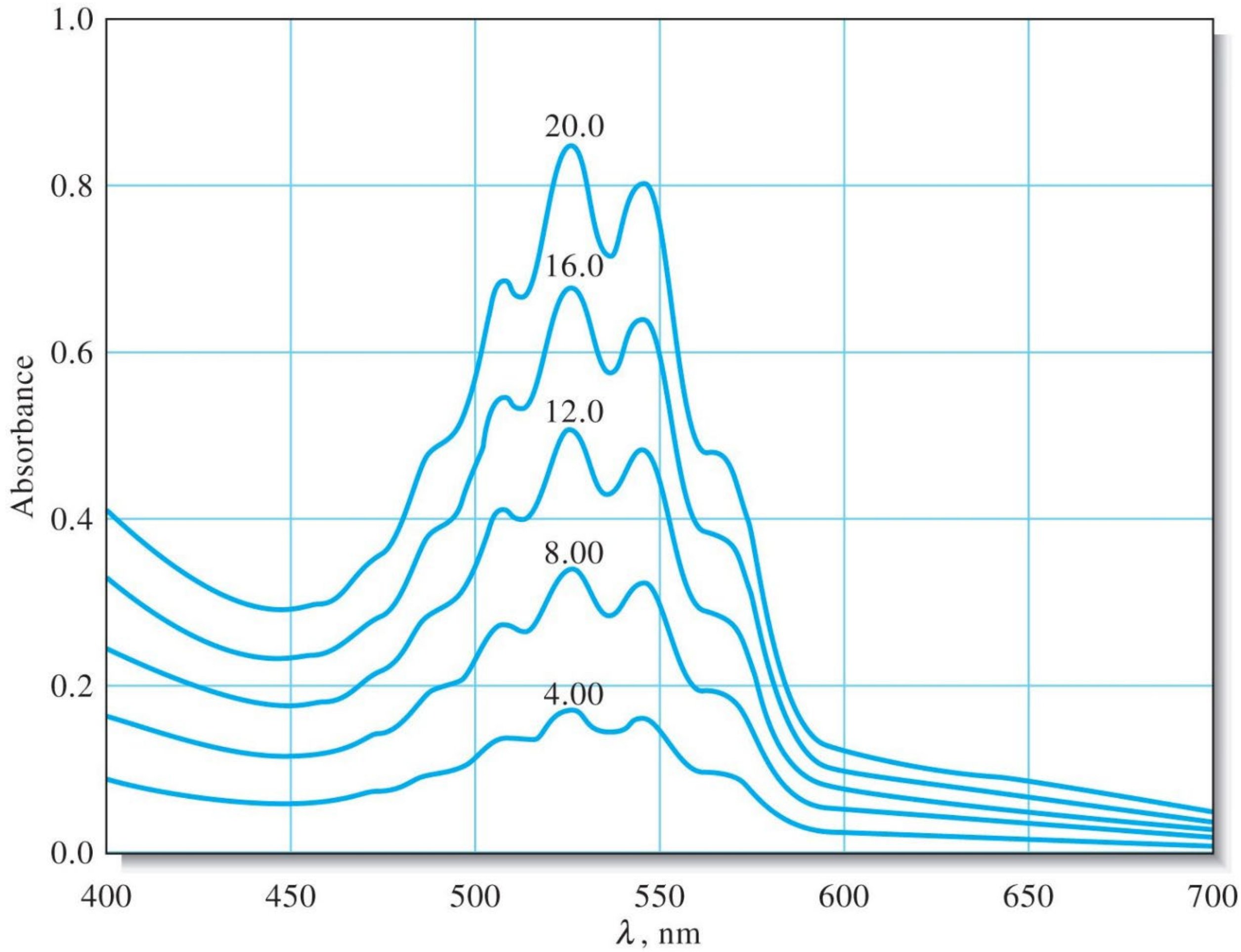
The Visible Spectrum

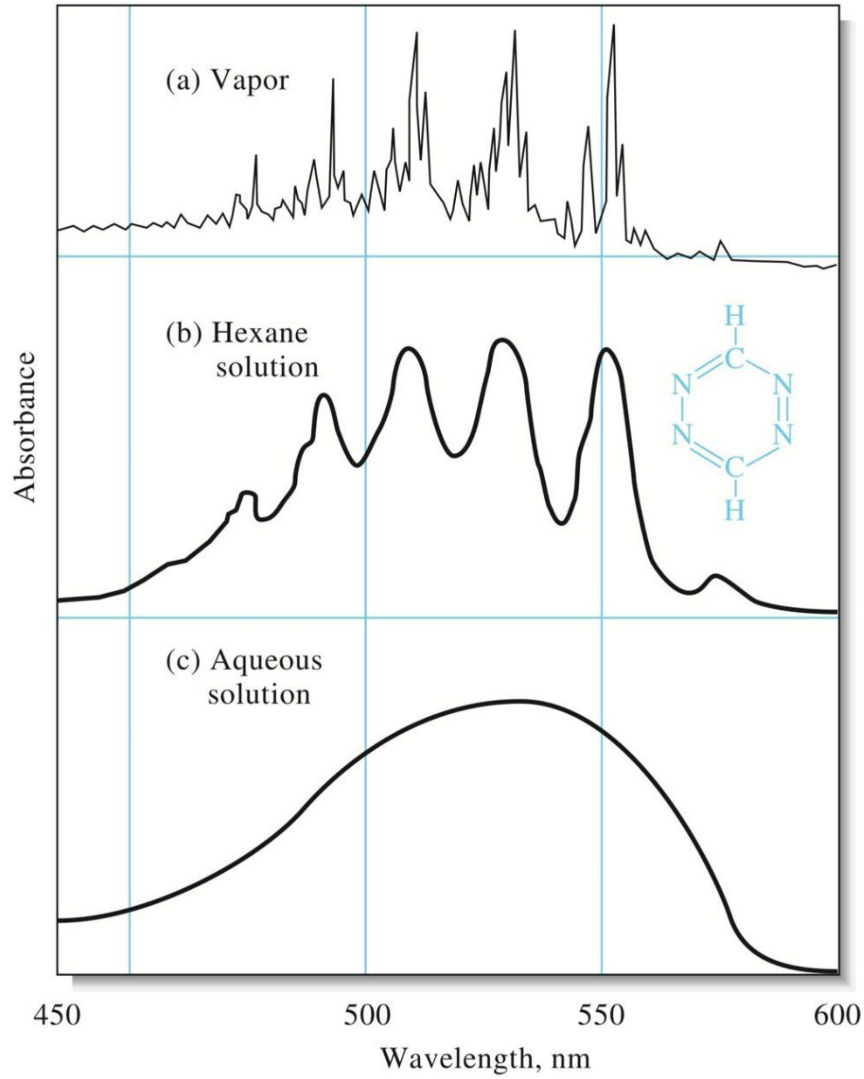
Wavelength Region Absorbed, nm	Color of Light Absorbed	Complementary Color Transmitted
400–435	Violet	Yellow-green
435–480	Blue	Yellow
480–490	Blue-green	Orange
490–500	Green-blue	Red
500–560	Green	Purple
560–580	Yellow-green	Violet
580–595	Yellow	Blue
595–650	Orange	Blue-green
650–750	Red	Green-blue

Figure 24F-4a p665

Absorption Spectra

An absorption spectrum is a plot of absorbance versus wavelength.





(Reproduced from S. F. Mason, *J. Chem. Soc.*, **1959**, 1263, **DOI**: 10.1039/JR9590001263, with permission of The Royal Society of Chemistry.)

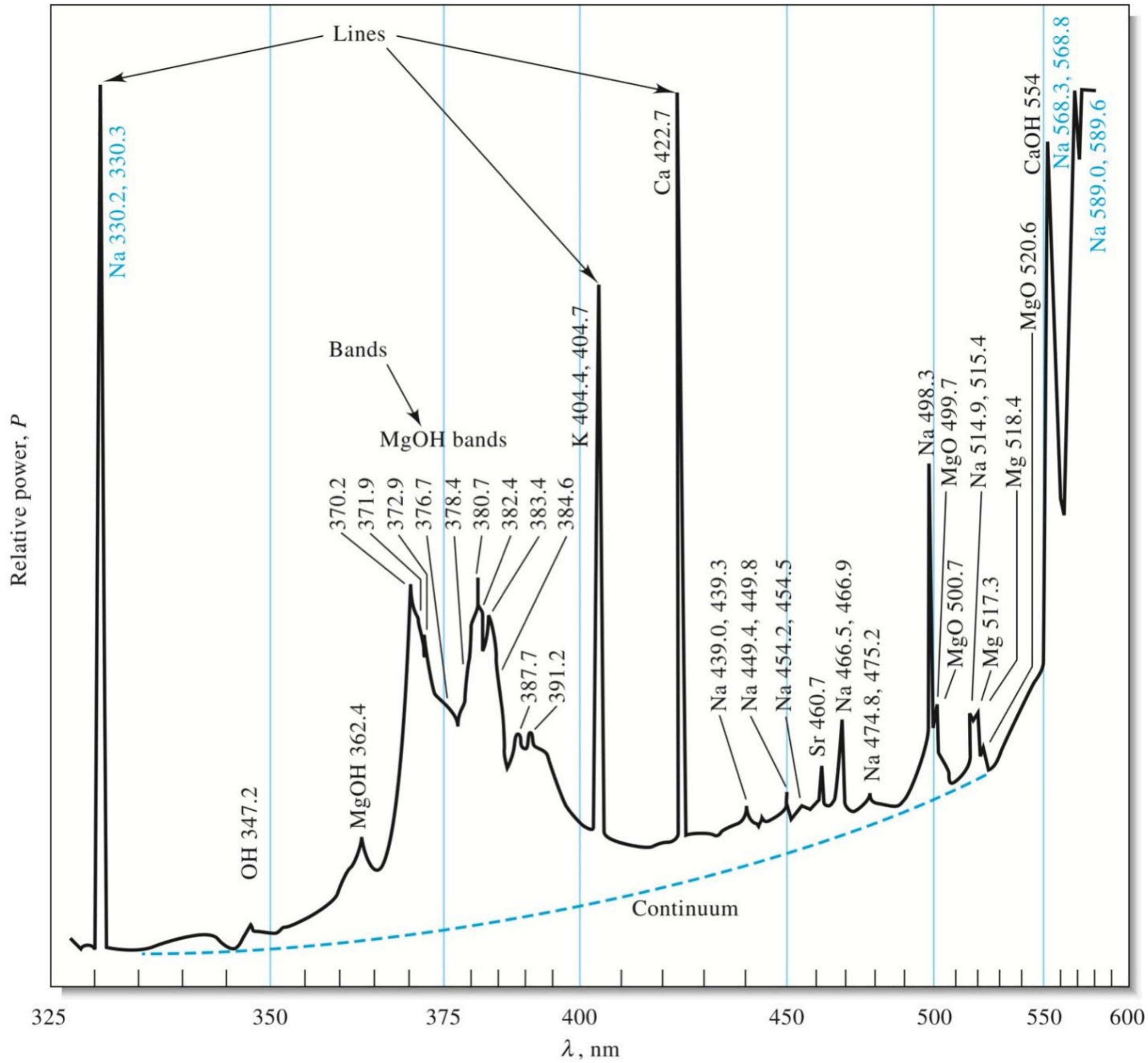
A) Separated in the gas state, enough that the molecule or atom can vibrate and rotate freely.

B) Loss of a degree of freedom and molecules cannot rotate freely.

C) Even stronger molecular forces, causes the spectrum to be smoothed out.

Emission of Electromagnetic Radiation

Emission spectra of brine sample within oxyhydrogen flame



(R. Hermann and C. T. J. Alkemade, *Chemical Analysis by Flame Photometry,* 2nd ed., New York: Interscience, 1979, p. 484.)

Three types of spectra are superimposed in the above figure

<u>Line Spectra</u>

Occur when the radiation species are individual atoms or ions that are well separated in the gas state.

<u>Band Spectra</u>

An emission band spectrum is made up of many closely spaced lines that are difficult to resolve.

Bands arise from the numerous quantized vibrational levels that are superimposed on the ground state electronic energy level of a molecule.

<u>Continuum Spectrum</u>

A spectral continuum has no line character is generally produced by heating solids to a high temperature.